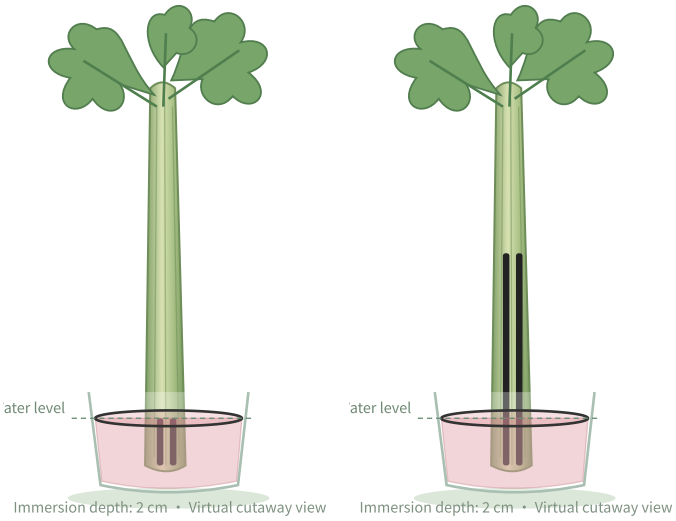
Name: _____ Class and student number: _____

The red water trace in celery

What the school gardening group found



After some time

Celery leaf stalk (petiole (葉柄))

This part looks like a stem. It contains xylem (木質部), so we can observe water moving upwards. It is a leaf stalk, called a petiole (葉柄).

This diagram shows an imaginary cutaway so you can see the water trace inside. The celery used in the experiment has not been cut lengthwise.

What can the red water trace tell us?

The gardening group puts celery with leaves in red water. They observe it before and after some time has passed.

Look at the two pictures on the left and record the changes you see. The group wants to know: do light and leaves affect water transport?

Main investigation question

When other conditions are the same, how does light intensity (光強度) affect the upward speed of the red water trace in celery with leaves?

Your first observations

What changes do you see? Describe them first. You do not need to explain why yet.

Complete your prediction

If light intensity (光強度) increases, the upward speed of the red water trace will （increase ／ decrease ／ show no clear change）.

Circle one answer in each set of brackets in the sentence.

My reason

Write your ideas. Your prediction does not have to match the expected result.

Identify the variables

Choose the factors that fit each variable type. Save your ideas now. You can view feedback after you submit.

Independent variable (the factor changed on purpose)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

Dependent variable (the result measured)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

Controlled variables (the factors kept the same)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

What assumptions does this investigation make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all the assumptions that apply. You may choose more than one.

- ☐ Changing the lamp distance does not make the setups have different temperatures.
- ☐ Setups A, B and C have the same total leaf area.
- ☐ Leaf area does not affect the results, so it does not need to be kept the same.
- ☐ Humidity (濕度) and air movement are the same for all setups.

The control group

How would you plan a comparison setup to find out whether leaves help with water transport?

(Hint: the control group provides a basis for comparison. It helps check whether the change in the experiment is caused by the independent variable rather than other external factors. The control group therefore does not receive the treatment being tested.)

(Hint: how is the control setup different from the experiment setup?)

Draw and label your four setups

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

Include the celery, leaves, red water, lamps, distances and measurement positions. Put 2 cm of each celery leaf stalk into the red water.


Celery ×4


Desk lamp ×4


Cup of red water ×4



30 cm ruler ×4


Scissors ×1



Timer ×1

Setup labels / written design

Add labels for your drawing, or describe all four setups in words instead of drawing them.

Save the original design before giving out B. Students keep a copy of A for reflection.

Carry out the investigation and collect data

All four setups run together. Take measurements every 10 minutes of simulated time.

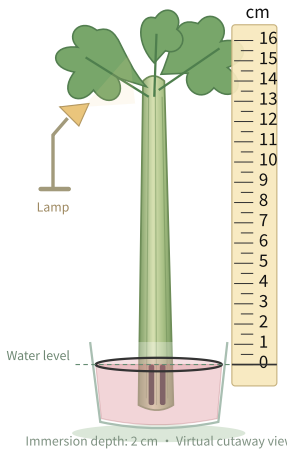
✦Conditions kept the same: Temperature of 25°C, the same humidity (濕度) and air flow, the same red water concentration, a depth of 2 cm in the water, and leaf stalks of similar length and thickness. Setups A, B and C have the same total leaf area. Setup D has no leaves.

In a real experiment: Keep the experiment and timer running while you measure. Read and record the data quickly at each planned time. Do not stop the timer to take measurements. This simulation pauses the picture to help you observe and record. The time you spend doing this is not part of the 30 simulated minutes.

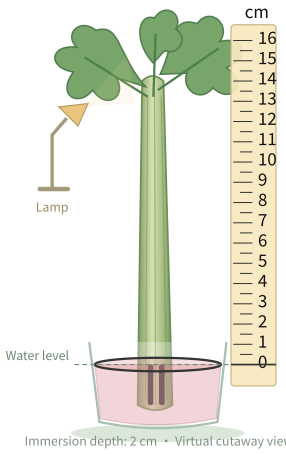
Read the rulers in the static images and record your own readings. These images use the same teaching model as the current VL.

0 min

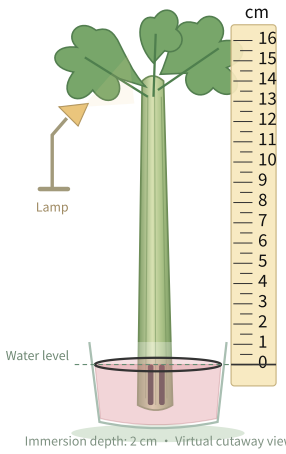
Setup A • With leaves
10 cm • 1200 lux



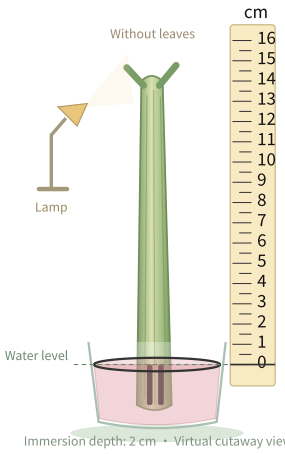
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Your measurement record

Simulation time (min)	Distance moved upwards by the red water trace (cm)			
min	Setup A: light distance 10 cm With leaves	Setup B: light distance 20 cm With leaves	Setup C: light distance 30 cm With leaves	Setup D: light distance 20 cm Without leaves
0				
10				
20				
30				

After entering a reading, press Enter to move to the next box. After Setup D, it moves to the confirm button. Your first readings are kept, and changes are also recorded. The platform does not replace your readings with reference values.

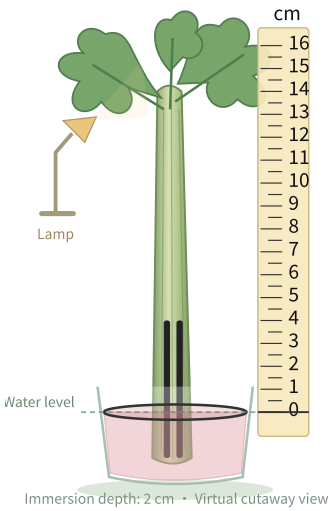
If you revise an answer, keep the first answer and write the change on the two lines below.

Carry out the investigation and collect data

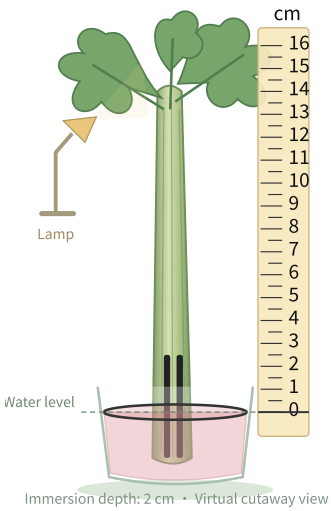
Read the rulers in the static images and record your own readings. These images use the same teaching model as the current VL.

10 min

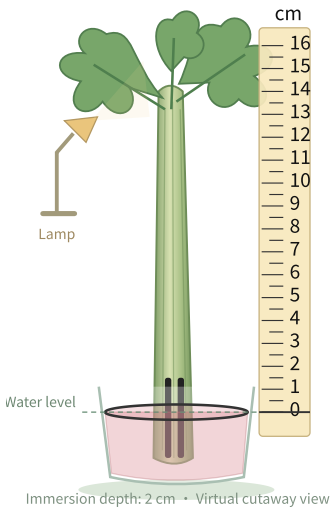
Setup A • With leaves
10 cm • 1200 lux



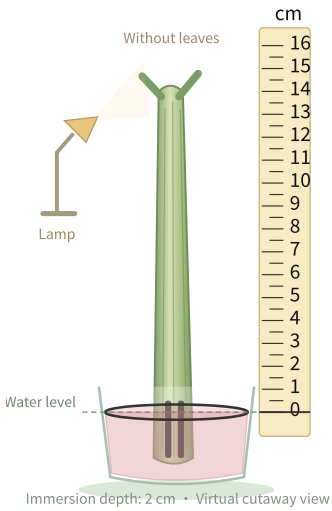
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux

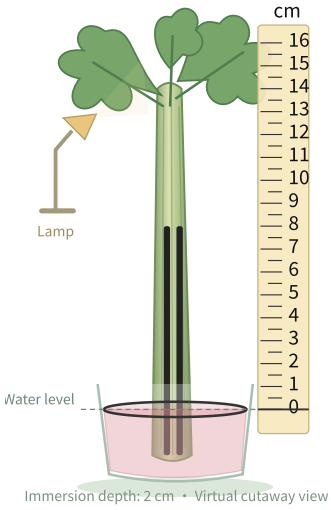


Setup D • Without leaves
20 cm • 300 lux

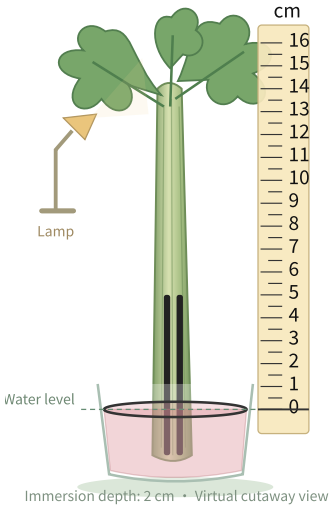


20 min

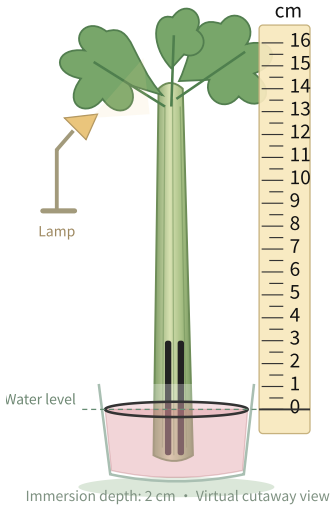
Setup A • With leaves
10 cm • 1200 lux



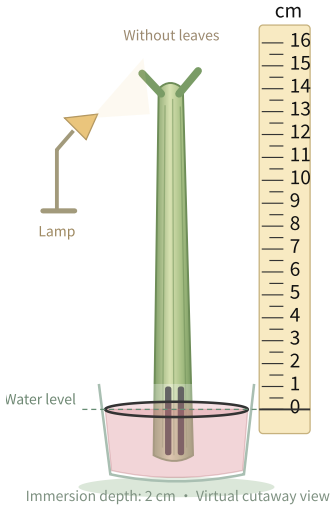
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux

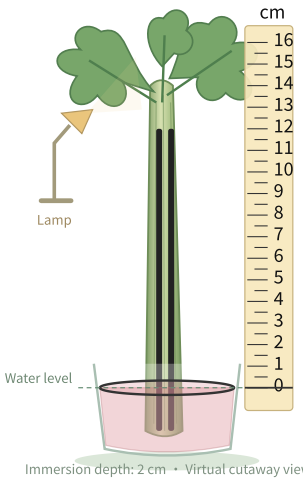


Drag the ruler left or right to place it beside the leaf stalk. Its 0 mark stays at the water surface, so you cannot move it up or down. Read the height of the top of the red water trace. Enter the reading to one decimal place. The arrows beside each reading change it by 0.5 cm.

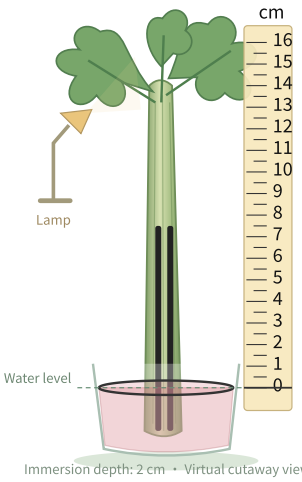
Each 10 minutes of simulated time takes about 3 seconds. This is a model for learning. It does not predict the exact results of a real experiment.

30 min

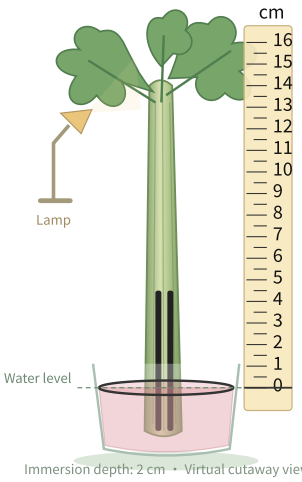
Setup A • With leaves
10 cm • 1200 lux



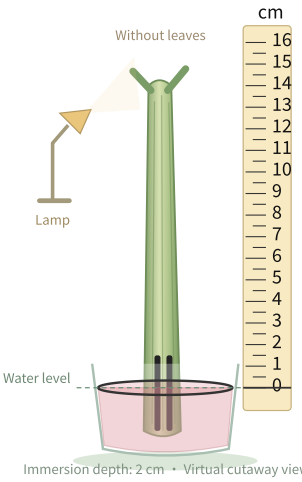
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Calculate the average upward speed

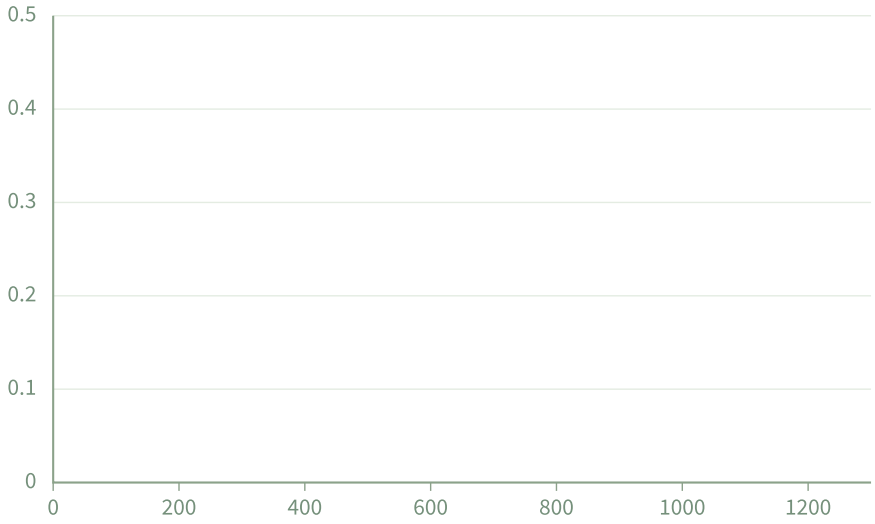
Use your readings and the time taken to calculate the average upward speed for each setup. Record each answer to two decimal places.

Average upward speed (cm/min) = distance moved upwards by the red water trace (cm) ÷ time taken (min)
Distance moved upwards = reading at 30 minutes − reading at 0 minutes. The time taken in this investigation is 30 minutes.

Setup	Average upward speed (cm/min)
Setup A • With leaves • 10 cm	
Setup B • With leaves • 20 cm	
Setup C • With leaves • 30 cm	
Setup D • Without leaves • 20 cm	

Light intensity (光強度) and the upward speed of the red water trace

Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

With leaves: Setups A, B and C | Without leaves: Setup D (a separate comparison point)

Move the pointer over the graph to see its (X, Y) coordinates (坐標). Choose a setup, then click the graph to plot its point. Click an existing point to edit its coordinates (坐標) in the boxes below. Select ‘Confirm / update point’ to save. You can also type coordinates (坐標) to plot a point. Record Y coordinates (坐標) to two decimal places. Join the points for setups with leaves with a smooth curve. Show Setup D separately.

Use your graph to reach a conclusion

Circle one answer in each set of brackets in the sentence.

1. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will （increase ／ decrease ／ show no clear change ／ show no consistent trend）.

2. What is the role of leaves?

At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is （faster ／ slower ／ similar）.

This comparison supports the idea that leaves （help water move upwards ／ stop water moving upwards ／ have no effect on water moving upwards）.

3. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?（The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). ／ The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). ／ If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).）

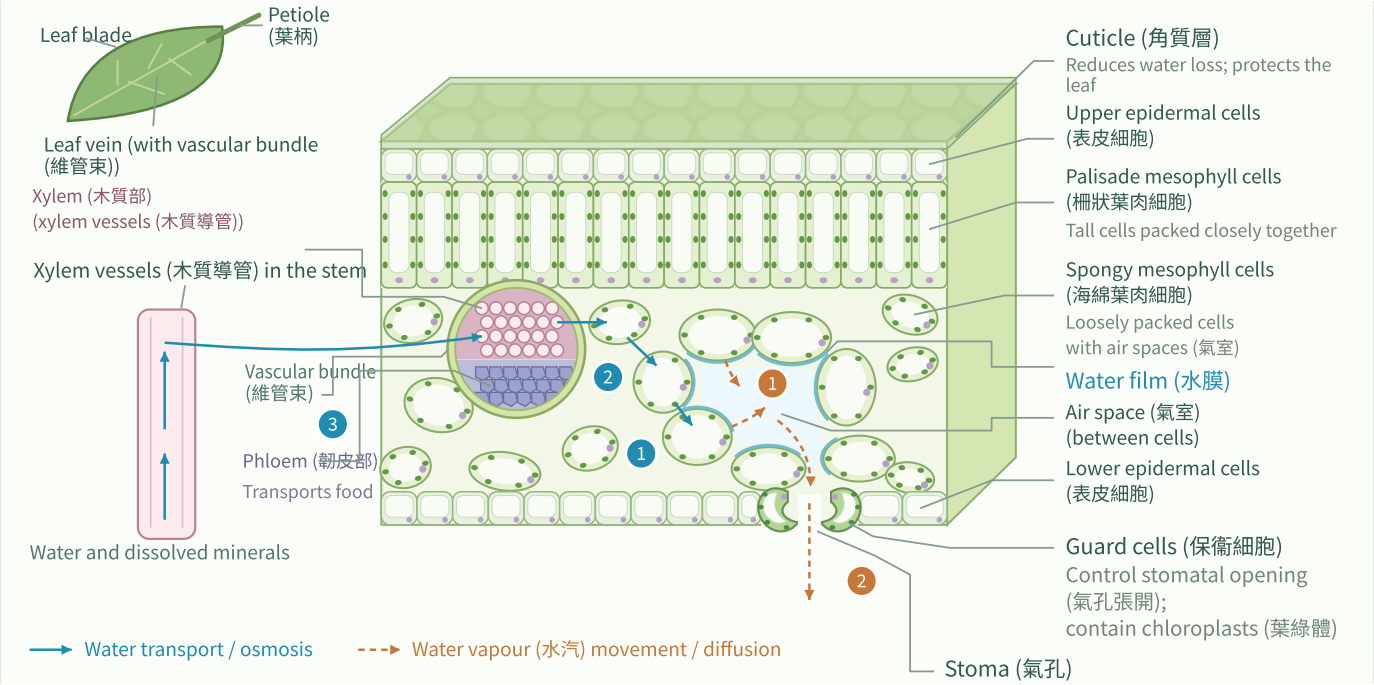
If you revise an answer, keep the first answer and write the change on the two lines below.

Give out C only after the original investigation answers are fixed.

Name: _____ Class and student number: _____

Give out C only after the original investigation answers are fixed.

Learn about transpiration (蒸騰) from the red water trace



Transpiration (蒸騰) through stomata (氣孔)

1 Water evaporates (蒸發) from the surfaces of mesophyll cells (葉肉細胞) into the air spaces (氣室).

2 Water vapour (水汽) in the air spaces (氣室) diffuses through stomata (氣孔) into the air.

How transpiration pull (蒸騰拉力) forms

1 Mesophyll cells (葉肉細胞) lose water, so their water potential falls.

2 Mesophyll cells (葉肉細胞) take in water from nearby cells by osmosis.

3 Water is drawn out of the xylem vessels (木質導管), creating transpiration pull (蒸騰拉力).

1. Transpiration (蒸騰) is the process in which water in a plant evaporates (蒸發) and is lost to the air as water vapour (水汽). Water vapour (水汽) mainly leaves through stomata (氣孔) in leaves and herbaceous stems (草本莖).
   a. Water evaporates (蒸發) from the water film (水膜) on the surface of mesophyll cells (葉肉細胞) and enters the air spaces (氣室) in the spongy mesophyll as water vapour (水汽). When the water vapour (水汽) concentration in the air spaces (氣室) is higher than in the air outside, water vapour (水汽) leaves the leaf through stomata (氣孔) by diffusion.
   b. When mesophyll cells (葉肉細胞) lose water, their water potential falls. They take in water from nearby cells by osmosis. This draws water out of the leaf's xylem vessels (木質導管) and creates transpiration pull (蒸騰拉力), which pulls water upwards through the stem's xylem vessels (木質導管).
2. Under suitable conditions, such as enough water being available, higher light intensity (光強度) usually makes stomata (氣孔) open wider. Water vapour (水汽) then leaves the leaf faster by diffusion, so the transpiration rate (蒸騰速率) increases.
   Note: More light may also make the leaf's temperature rise, increasing transpiration (蒸騰) further. This investigation therefore assumes that all setups have the same temperature so we can compare the effect of light intensity (光強度).
3. Transpiration pull (蒸騰拉力) keeps roots taking in water and moves water and dissolved minerals absorbed by the roots upwards through xylem vessels (木質導管) to the leaves.

See textbook section 11.1, pages 11-3 to 11-8. The red water trace gives evidence of water transport. Its upward speed is not a direct measure of the transpiration rate (蒸騰速率). Comparing celery with and without leaves alone also cannot prove the role of stomata (氣孔).

Internal leaf structure and transpiration (蒸騰)

Orange numbers: transpiration (蒸騰) through stomata (氣孔) | Blue numbers: how transpiration pull (蒸騰拉力) forms

Enlarged view, not to scale. The celery stalk in this experiment is a petiole (葉柄). The stem shown here explains transport through xylem vessels (木質導管).

Return to your first ideas

Refer to the original hypothesis and reason in A. You do not need to rewrite them from memory.

Do your data support your original hypothesis? Use ideas from the key learning points above to improve your original explanation. Explain how it relates to your data.

For teachers only. Do not give this part to students.

Distribution and evidence preservation

1. Following proposal 4.1 and 4.3.2, these worksheets use the same scenarios, concepts, prompts and static data, before formal teaching. Keep weekly worksheets separate from the formal 10-item SPSAI and 4-item CKT pre- and post-tests.
2. Give out A first. Save the original observation, hypothesis, reason, variables, assumptions and setup design. Keep a copy students can consult during reflection.
3. Give out B after saving the original design. Static images replace interaction for the same four setups and four times. Students complete the readings, calculations, graph and conclusions.
4. Keep first readings legible and record revisions separately. Give out C only after the original investigation answers are fixed; do not reveal the learning points earlier.
5. Students consult their original hypothesis and reason in A and complete the same reflection in C. D and the full pack are for teachers only; distribute the parts separately.
6. Use the same situation, questions, hints, data, language and available time for both groups. Static reading and digital interaction are different treatments; paper completion is not a measure of operating skill.
7. The current VL2 has no extension investigation; the paper does not add one. Keep formal research pre- and post-tests separate from these activity worksheets.
8. Test-print B and C in grayscale before use. Check liquid surfaces, water traces, rulers and scientific labels. Time, clicks and completion rate do not directly become SPS scores.

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

Blank teacher scores mean awaiting marking, not zero. The six SPS areas total 24 points; new knowledge totals 8.

For teachers only. Do not give this part to students.

Reference answers

Independent variable	Light intensity (光強度)
Dependent variable	Height / average upward speed of the red water trace
Controlled variables	Temperature / Humidity (濕度) and air movement / Red water concentration and depth of immersion (浸入深度) / Total leaf area of the setups with leaves / Length and thickness of the celery stalks
What assumptions does this investigation make?	Changing the lamp distance does not make the setups have different temperatures. Setups A, B and C have the same total leaf area. Humidity (濕度) and air movement are the same for all setups.
1. How is light intensity (光強度) related to the upward speed of the water trace?	increase
2. What is the role of leaves?	faster
2. What is the role of leaves?	help water move upwards
3. What can this experiment tell us?	The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率).

min	A (cm)	B (cm)	C (cm)	D (cm)
0	0.0	0.0	0.0	0.0
10	4.0	2.5	1.5	0.5
20	8.0	5.0	3.0	1.0
30	12.0	7.5	4.5	1.5
cm/min	0.40	0.25	0.15	0.05

For teachers only. Do not give this part to students.

References for written answers and assumptions

After some time, the red water trace extends upwards in some parts inside the celery stalk.

There is no single answer. Your hypothesis should be testable (可測試的) by changing light intensity (光強度) and measuring the upward speed of the water trace.

Explain the link you expect between light and water transport. Your prediction does not have to match the final result.

Compare setup D without leaves with setup B with leaves. Both have a lamp distance of 20 cm and the same other conditions. The main difference is whether leaves are present.

Materials: 4 celery stalks, 4 desk lamps, 4 cups of red water, 4 rulers of 30 cm, 1 pair of scissors and 1 timer. Set the lamp distances for the three celery stalks with leaves to 10, 20 and 30 cm. Use the same lamp distance for the setup without leaves and one setup with leaves. Immerse each stalk 2 cm into the red water.

Changing the lamp distance does not make the setups have different temperatures.

Temperature can also affect transpiration (蒸騰). This simulation keeps all setups at 25°C so light and temperature do not change together.

Setups A, B and C have the same total leaf area.

Leaf area can affect water loss, so setups A, B and C with leaves should have the same total leaf area. Setup D without leaves provides a separate comparison.

Leaf area does not affect the results, so it does not need to be kept the same.

Leaf area can affect water loss. Keep it the same when comparing light intensity (光強度).

Humidity (濕度) and air movement are the same for all setups.

Humidity (濕度) and air movement can also affect transpiration (蒸騰). Keep them the same when comparing light intensity (光強度).

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
First observations	2	Teacher marks	Describe the red trace extending upwards in parts of the leaf stalk; separate observation from explanation.	1: Mention the trace or colour change but omit its position or direction.	No answer, unrelated answer, or report an unobserved mechanism as a direct observation.
Ruler readings	2	Automatic	All 16 final confirmed readings are within 0.25 cm of the ruler model.	2 × number of correct readings ÷ 16, to two decimal places.	No correct confirmed readings.
Variable classification	4	Automatic	Independent: light intensity (光強度) (1); dependent: trace height / average upward speed (1); five controlled variables (2).	Controlled variables: 0.4 each. Selecting the independent or dependent variable as controlled gives zero for that column.	Not answered or incorrectly classified.
Testable (可測試的) hypothesis and reason	2	Teacher marks	Predict how changing light intensity (光強度) affects trace speed and give a relevant reason. The prediction need not match the result.	1: A testable (可測試的) hypothesis with an incomplete or unclear reason.	No testable (可測試的) hypothesis, or copy a conclusion without a prediction made before the experiment.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

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Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Experiment assumptions	1	Automatic	Select only unchanged temperature, equal total leaf area in the leafy setups, and equal humidity (濕度) and air flow.	Credit requires the whole set to be correct; no partial credit.	An extra, missing or absent selection.
Control setup design	1	Teacher marks	Compare with and without leaves at the same light distance, keeping other conditions the same.	This item receives 0 or 1 only.	Other factors are not kept the same, or no answer.
Four complete measurement rounds	2	Automatic	Valid confirmed readings for all four setups at 0, 10, 20 and 30 min.	0.5 per complete round. Reading accuracy is scored under observing, not counted twice.	No complete confirmed measurement rounds.
Experiment setup plan	2	Teacher marks	Drawing or words show four stalks: three leafy light distances and a leafless control at the same distance; include red water and ruler / timing arrangements.	1: A workable plan missing one main arrangement.	Unworkable, unrelated or incomplete.

Automatic scoring in the shared guide refers to VL Excel. Teachers mark paper answers using the same criteria; handwriting is not read automatically.

For teachers only. Do not give this part to students.

Scoring guide

The activity scoring guide is a draft for researcher calibration, not a validated research instrument.

Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Average upward speed calculations	2	Automatic	For each setup: (30 min reading − 0 min reading) ÷ 30. Each correct setup earns 0.5.	Use the student’s final readings. New records use two decimal places and a 0.0051 cm/min rounding tolerance; old records retain three decimal places and 0.00051 tolerance.	No correct calculations. Do not substitute model readings when assessing arithmetic.
Four conclusion choices	2	Automatic	Greater light intensity (光強度) usually gives faster movement; B/D comparison matches student calculations; leaves help transport; the trace is indirect evidence.	0.5 per correct choice. The B/D comparison needs sufficient data.	Incorrect or unanswered.
Graph points and curve	2	Automatic	Four points: X = each setup’s lux, Y = the student’s calculated speed (0.375 each); correctly join A/B/C (0.5).	Calculate with unrounded point credit, then round to two decimals. Do not include D in the leafy curve.	No correct points or curve. Calculation errors are scored separately; assess the graph using the student’s calculations.
Written communication	2	Teacher marks	Observations, setup plan and reflection communicate ideas clearly, using clear data, setup names and units. Assess expression separately from conceptual accuracy.	1: Understandable but vague or missing necessary labels.	Cannot be understood or no assessable communication.

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Reflection: transpiration (蒸騰) and stomata (氣孔)	2	Teacher marks	The revised explanation correctly describes evaporation (蒸發) from mesophyll water films (水膜) and water vapour (水汽) leaving through stomata (氣孔) by diffusion.	1: Correctly mention leaf evaporation (蒸發) or vapour loss through stomata (氣孔), but the mechanism is incomplete.	No use of the concept or a core misconception.
Reflection: xylem (木質部) and transpiration pull (蒸騰拉力)	2	Teacher marks	Link transpiration (蒸騰) to transpiration pull (蒸騰拉力) and upward water transport through xylem vessels (木質導管).	1: Correctly mention xylem (木質部) transport or transpiration pull (蒸騰拉力) without linking them.	No use of the concept or a core misconception.
Reflection: effect of light intensity (光強度)	2	Teacher marks	Link stronger light to wider stomatal opening (氣孔張開), transpiration (蒸騰) and upward transport. Do not equate trace speed with transpiration rate (蒸騰速率).	1: Correctly describe stronger light and faster movement but omit the mechanism.	Reverse the relationship, directly equate trace speed with transpiration rate (蒸騰速率), or do not use the concept.
Reflection: revising the original explanation	2	Teacher marks	Judge whether the original hypothesis is supported by the student’ s data, then revise or improve the explanation using data and learning points. A correct initial hypothesis can earn full credit.	1: A judgement or revision without a clear data or concept link.	Only copy learning points, do not test the original hypothesis, or do not submit reflection.

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Category / item	Maximum	Scoring method	Full credit	Partial credit	Zero credit
Totals and completion requirements	32	Formula totals	Six SPS areas × 4 = 24; four knowledge areas × 2 = 8; total 32. Show the overall score only after all nine teacher scoring cells and the reflection are complete.	Blank means awaiting marking, not zero. Excel recalculates on opening. Save the workbook after marking.	A new download does not import scores from an earlier workbook. Teacher scores are kept only in the saved Excel file.
Colours and research use	0	Information	Background colours distinguish SPS and knowledge. Green = correct / full credit; red = incorrect / zero; orange = partial; awaiting marking uses a neutral colour.	Written-answer cells change colour when marked. PDFs show no scores; activity records are not SPS scores.	This is a VL2 draft. Align criteria, weights and raters across the four modules before longitudinal comparisons; it is not a validated SPSAI replacement.

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